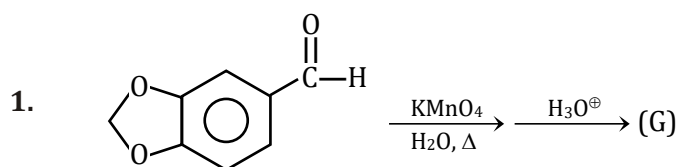


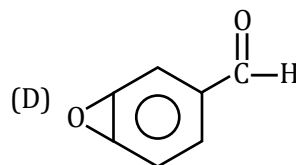
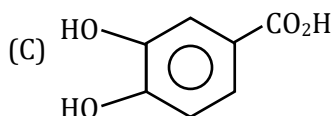
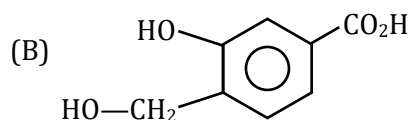
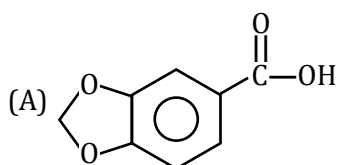


Oxidation : RACE-01

Oxidation of alcohol by Strong oxidising agent (KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, CrO_3 , H_2SO_4)

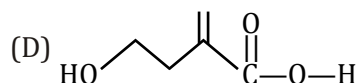
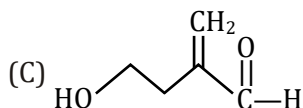
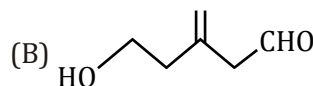
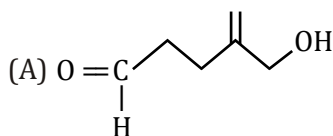
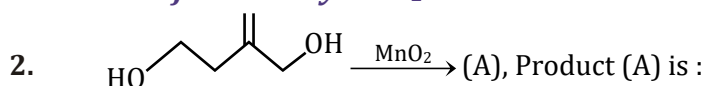


Product (G) is :



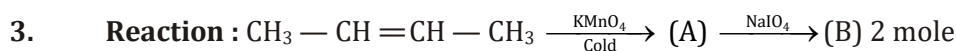
COX105

Oxidation of alcohol by MnO_2

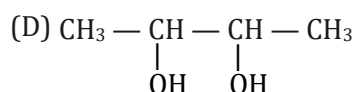


COX103

Oxidation of diol by HIO_4 and $\text{Pb}(\text{OAc})_4$



Product (B) is :



COX104

Oxidation

Oxidation of Alkenes (Baeyer's Reagent, OsO₄, Hot KMnO₄, Acidic KMnO₄, Peroxy Acid followed by hydrolysis, Ag/O₂ followed by hydrolysis)

4. $\text{CH}_3\text{-CH=CH-CH}_3 \xrightarrow{\text{X}}$ Product is Y (non-resolvable) then X can be –
 (cis-2-butene)
 (A) Br₂ water (B) HCO₃H/H₃O⁺
 (C) Cold alkaline KMnO₄ (D) All of these

COX116

5. Oxidation of isobutylene with hot acidified potassium permanganate gives:
 (A) Acetone + CO₂ (B) Acetic acid
 (C) Acetic acid + CO₂ (D) Acetic acid + acetone

COX112

6. During Epoxide formation, the peroxy-acid acts as an :-
 (A) Nucleophile (B) Electrophile (C) Base (D) Acid

COX113

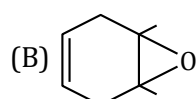
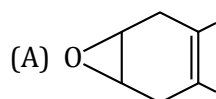
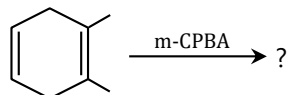
7. $\text{CH}_3\text{-C}\equiv\text{C-CH}_3 \xrightarrow[\text{S-quinoline}]{\text{H}_2, \text{Pd-CaCO}_3} \text{(A)} \xrightarrow[\text{H}_2\text{O/H}^+]{\text{m-CPBA}} \text{Products}$

Product contains :-

- (A) Erythro Racemic (B) Threo Racemic
 (C) Meso Compound (D) Diastereomers

COX114

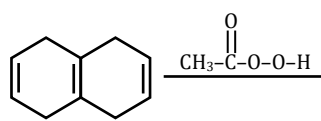
8. Identify the major product for the following reaction :-

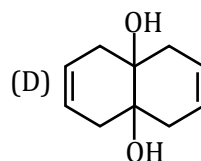
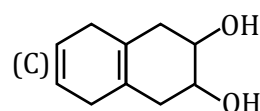
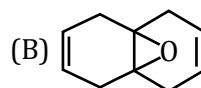
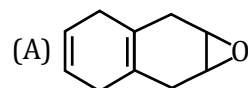


(C) Both (A) and (B)

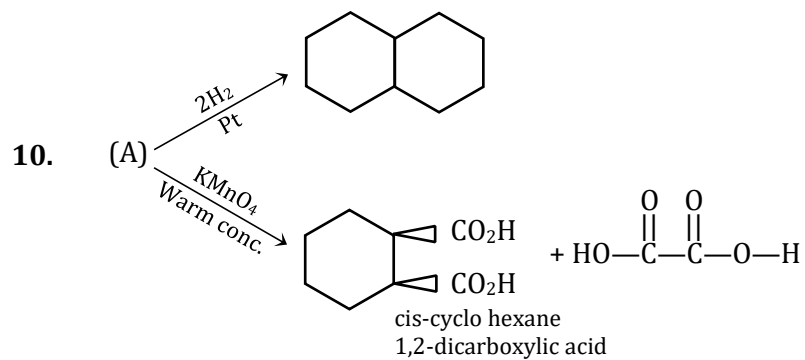
(D) None of these

COX115

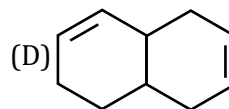
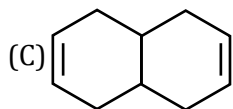
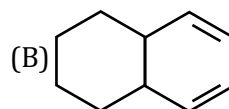
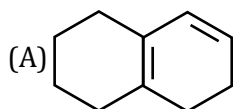
9.  (A) ; Product (A) is :
 (major)



COX102



Identify (A).



COX106

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	C	A	C	A	B	B	B	B	B